

Design Rulebook

Project Goals

Generate 3D printable electromagnetic circuits in OpenSCAD. Circuits will consist of a protopasta conductive PLA trace layer and an insulating layer that separates the traces.

Physical Laws

- Conductive trace layer resistivity (ρ): 15 $\Omega \cdot \text{cm}$.
- Resistance: $R = \frac{\rho A_c}{L}$.
- A_c = cross-sectional area of material.
- L = length of material.

Design Principles

- Conductive trace width: 2mm.
- Conductive trace spacing: 2mm.
- Conductive trace height: 2mm.
- Minimum Corner Angle: 45 degrees.
- Insulating layer thickness: 2mm.
- Insulating layer and Conductive Trace layer are generated separately, not as a single OpenSCAD project.
- The insulating layer should provide 2mm of clearance on each trace boundary
 - i.e., the trace footprint outline is 25.4 mm by 25.4 mm; the insulating layer footprint should be 29.4 mm by 29.4 mm, matching the general shape of the trace footprint outline.
- 4 mm by 4 mm contact pads should be placed at the end of each circuit, with full contact between the end of the trace and the contact pad.

Optimization Goals

- Use the length and routing needed to use the minimum amount of materials, while accounting for the Design Principles and Physical Laws.

Style Preferences

- The insulating layer serves as a flat platform for conductive traces to be printed on.
- Only create walls in the insulating layer if necessary to isolate distinct trace planes.

- Prefer serpentine-based resistors.
- Prefer interdigitated-based capacitors.